

Product List Block

Summary

Product List component displays product catalogs with pricing on web pages, allowing users to view and purchase products.

References Page Url

T [Celleste Image Analysis Software | Thermo Fisher Scientific - US](#)

T [EVOS LED Light Cubes | Thermo Fisher Scientific - US](#)

T [Molecular Biology Bulk Reagents and Lab Supplies for Sample Preparation | Thermo Fisher Scientific - US](#)

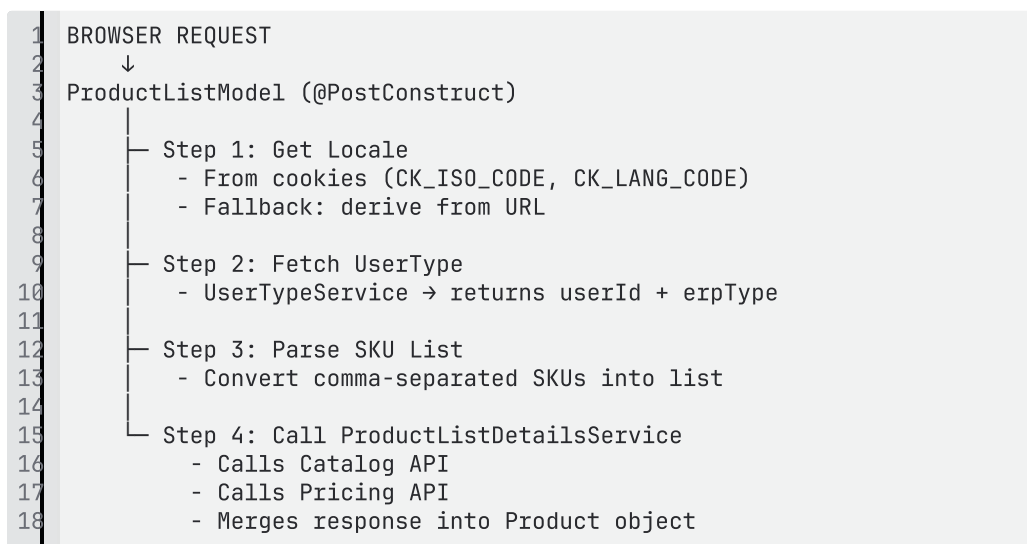
T [Transfected Cell Analysis | Thermo Fisher Scientific - US](#)

Current State

The Product List component is a flexible, data-driven AEM component that renders product information in a table format. It:

- Fetches product details via **Catalog Microservice API** based on SKU list
- Retrieves pricing via **Pricing Access API** with user-specific pricing
- Renders a dynamic HTML table with configurable columns (SKU, Name, Size, Price, Quantity)
- Supports **Add-to-Cart** via client-side JavaScript
- Handles multiple price access types (RequestQuote, OrderNow, NoPrice, etc.)

Data Flow



```
19
20     ↓
21 JSP renders product table
22     ↓
23 BROWSER DISPLAY
```

Service Layer Responsibility

`ProductListDetailsServiceImpl` performs:

- UserType retrieval
- OAuth token generation via `AuthTokenProvider`
- Catalog API call
- Pricing API call
- Consolidation into a single Product response

Frontend Responsibility

- Only renders data provided by backend
- Handles Add-to-Cart:
 - Validates quantity
 - Calls mini-cart
- No API calls from browser

EDS Approach

EDS does not support AEM-style server-side rendering. The published HTML is static.

To support dynamic product data while keeping SEO and security intact, a **two-layer architecture** is proposed:

1. Edge Worker (Presentation Layer)

- Reads product-list block from page
- Extracts SKU list and configuration
- Calls Product Microservice
- Renders HTML using returned JSON
- Injects final HTML into response

2. Product Microservice (Data Layer)

- Handles OAuth token generation
- Resolves user context (UserType)

- Calls:
 - Catalog API
 - Pricing API
 - Merges responses
 - Returns normalized JSON
-

Product List Block (Authoring in EDS)

The EDS Product List block will allow authors to configure:

- **SKU list** (comma-separated or structured input)
- **Column selection**, such as:
 - Size
 - Price
 - Quantity
 - Add-to-Cart

This ensures flexibility similar to the current AEM component while keeping authoring simple.

Important Design Decision

- Business logic **is not recommended to be implemented in Edge Worker**
- Edge should remain thin (rendering only)
- Microservice should handle all backend logic

Microservice can be implemented using:

- App Builder
 - AWS Lambda
 - Node.js service
-

Suggested Architecture View

```
1 Universal Editor
2   ↓
3 EDS Page (Product List Block)
4   ↓
5 Edge Worker
6   - Read SKUs
7   - Call Microservice
8   - Render HTML
9   ↓
10 Product Microservice
11  - Auth Token
```

```
12 - UserType
13 - Catalog API
14 - Pricing API
15 - Merge response
16   ↓
17 Backend APIs
```

Additional Notes

- Add-to-Cart remains **client-side** (same as current)
- **TFS is also planning to introduce a microservice for fetching product details.**
 - Need to confirm:
 - Scope of that service
 - Integration approach with EDS

Recommendation

Adopt **Edge Worker + Product Microservice pattern.**

- Keeps API logic secure and outside browser
- Maintains SEO via server-side rendering at edge
- Keeps edge layer simple and maintainable
- Reuses existing backend logic effectively

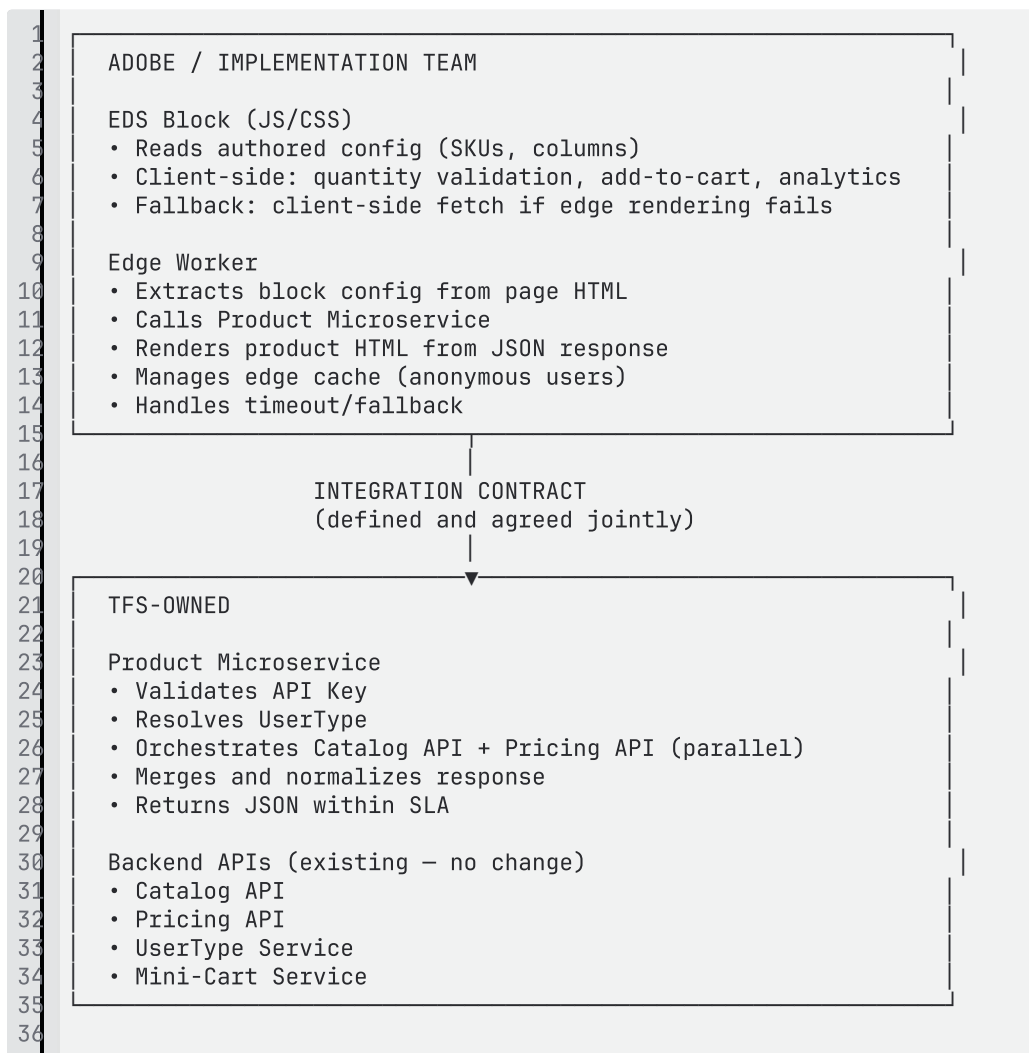
Runtime Ownership & Integration Boundaries

Ownership Matrix

Component	Owner	Responsibilities
Product List Block (EDS)	Adobe / Implementation Team	Block JS/CSS, UE authoring model, client-side rendering (fallback), quantity validation, analytics event push, add-to-cart UX
Edge Worker	Adobe / Implementation Team	Extract SKU/column config from page HTML, call Product Microservice, render product HTML table, inject into

		response, manage cache, handle timeout/fallback
Product Microservice	TFS	UserType resolution, Catalog API calls, Pricing API calls, response merging, normalized JSON response, SLA ownership
Mini-Cart Service	TFS (existing backend)	Cart operations, stock validation, order management

Integration Boundaries



Integration Contract

The integration contract between the Edge Worker and the Product Microservice will be defined and agreed jointly during implementation.

Aspect	Specification
Request	SKU list + locale/country + user context (cookie/token if authenticated)
Response	Normalized JSON — array of product objects with name, SKU, size, price, availability, price access type
Authentication	API Key in request header (X-API-Key)
SLA Target	< 500ms (p95 response time) for typical request (5–20 SKUs)
Timeout	2 seconds at the Edge Worker — if microservice does not respond within 2s, fallback is triggered
Method	POST (SKU list in request body) or GET (SKUs as query parameter) — to be agreed

Cache Rules:

- **Anonymous users:** Edge-cached product data with short TTL (5–15 minutes). Price changes reflect within TTL window.
- **Logged-in users:** All caches bypassed. Every request fetches live, personalized pricing from microservice. No edge caching for authenticated requests.
- **Stale-while-revalidate:** If microservice is temporarily unavailable, serve last cached response (anonymous users only) to avoid complete failure.
- **Final TTL is a business decision** — trade-off between performance (longer TTL) and pricing freshness (shorter TTL). Will be agreed during implementation.

Note: Product data (names, sizes, pricing, availability) is NOT authored. It is fetched dynamically at runtime from the Product Microservice based on the SKU list configured by the author. This block defines WHAT to show (SKUs + columns). The Edge Worker + Microservice handle HOW to render it.

UE Authoring Contract

AEM Resource

Property	Value
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Resource Type	core/franklin/components/block/v1/block
Block Name	Product List
Model ID	product-list
Block Type	Simple (flattened — no child items)

Editable Properties in Universal Editor

JCR Property	UE Properties Panel	Field Type	Required	Default	Notes
skuList	SKU List	text (textarea)	Yes	""	Comma-separated SKU IDs (e.g., "16096040,15596018,A33251,17909")
showSize	Size	boolean	No	true	Show/hide Size column
showPrice	List Price	boolean	No	true	Show/hide Price column
showQuantity	Quantity	boolean	No	true	Show/hide Quantity input column
showAddToCart	Add to Cart	boolean	No	true	Show/hide Add to Cart button column
showPdpLink	PDP UI	boolean	No	false	Show/hide link to Product Detail Page

Container vs Leaf Fields

The Product List is a **leaf block (flattened)**. All fields are on the block itself. No child items.

The product rows (individual products) are NOT authored — they are dynamically rendered at runtime based on the SKU list. The block is purely a **configuration block** — it tells the system what to fetch and which columns to display.

Block Registration

```
1 {
2   "title": "Product List",
3   "id": "product-list",
4   "plugins": {
5     "xwalk": {
6       "page": {
7         "resourceType": "core/franklin/components/block/v1/block",
8         "template": {
9           "name": "Product List",
10          "model": "product-list"
11        }
12      }
13    }
14  }
15 }
```

Component Model Definition

Field Definitions

Field	Component	Label	Required	Default	Authored / Derived
skuList	text	SKU List	Yes	""	Authored
showSize	boolean	Size	No	true	Authored
showPrice	boolean	List Price	No	true	Authored
showQuantity	boolean	Quantity	No	true	Authored
showAddToCart	boolean	Add to Cart	No	true	Authored
showPdpLink	boolean	PDP UI	No	false	Authored

Authoring Strategy

Classification: Flattened Block

Pattern	Reason
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Flattened block

Configuration-only block. Author provides SKU list + column preferences. No child items, no repeating elements. Product rows are rendered dynamically, not authored.